Multiplication theorem (Independent events):

If A & B are Independent of some sample space then,

$$P(A \cap B) = P(A) \cdot P(B)$$

$\cap$ indicates the A & B events occur together

Q. If a bag contains 8 white, 10 black ball

Conditional Event: If A & B are any two events of a sample space and if A occurred after the occurrence of B then it is called conditional event of A given B = (A/B).

Similarly, conditional event of B given A = (B/A)

Conditional Probability: (CP)

The conditional p. of event A when the B is already happened is denoted by P(A/B) & is defined by

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

(or) $$P(A \cap B) = P(A/B)\,P(B)$$

Similarly CP of B when A already happened

$$P(A \cap B) = P(B/A)\,P(A)$$

Bayes Theorem

$E_1, E_2 \cdots E_n$ are mutually exclusive and exhaustive events such that $P(E_i) > 0 \quad (i = 1, 2, \cdots n)$ in a sample space S and A is any other event in S intersecting every event E_i (ie, A can only occur with combination of any of $E_1, E_2 \cdots E_n$) such that $P(A) > 0$

If E_K is any event of $E_1 E_2 \cdots E_n$ where, $P(E_1), P(E_2) \cdots P(E_n)$ &

$P(A/E_1), P(A/E_2) \cdots P(A/E_n)$ are known then,

$$P(E_K/A) = \frac{P(E_K) \times P(A/E_K)}{P(E_1)P(A/E_1) + \cdots P(E_n)P(A/E_K)}$$

$$\boxed{P(E_K/A) = \frac{P(E_K)\, P(A/E_K)}{\sum_{i=1}^{n} P(E_i)\, P(A/E_i)}}$$

Q- In a bolt factory machines A, B, C

Random Variable (RV):

A variable whose value is defined from the outcomes of Random experiment

Ex1:- Tossing 2 coins

$S = \{HH, HT, TH, TT\}$

X = no of tails

Ex2:- Rolling a die

$S = \{1, 2, 3, 4, 5, 6\}$

X = Perfect square

There are two types.

* A random variable which accepts only integers is called Discrete RV.
* A random variable which accepts both integers & Fractions is called Continuous RV

Discrete RV:

→ **Prob. Function**: For a DRV the Prob. function is $P(X=x)$ and it finds Prob. for each x value

→ **Prob. Distribution (or) Mass Function**

Let X = RV whose values are $x_1, x_2, \ldots x_n$ with Probabilities $p_1, p_2, \ldots p_n$.

The Prob. Distribution can be written as

X	x_1	x_2	----	x_n
$P(X)$	p_1	p_2	----	p_n

Mean $\mu = \sum p_i x_i$

Expetation $E(X) = \sum p_i x_i$

Variance $\sigma^2 = \sum p_i x_i^2 - \mu^2$

Q. Two dice are thrown. X is a RV such that $X(a,b) = \max(a,b)$

Probability Density Function

Suppose X is a continuous R V

$f(x)$ = P. Density Function and corresponding Prob. are given by

$$\int_a^b f(x)\,dx$$

Properties

i, f is always +ve, $f(x) \geq 0$

ii, Total P = 1

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

Suppose X is a continuous RV then

$$\text{Mean } \mu = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$\text{Variance } \sigma^2 = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$$

Let Median $= m$

$$\int_a^m f(x)\,dx = \int_m^b f(x)\,dx = \frac{1}{2}$$

Suppose X is a continuous RV, F(x) is called Distribution function. It calculates Cumulative Probability defined as

$$F(x) = \int_{-\infty}^{x} f(x)\,dx$$

$$\Rightarrow F'(x) = f(x)$$

Suppose f(x) is PDF & F(x) is Distributions function, then.

Expected f

	Expected
34	freq
44	$= P(X=x_i) \times 100$
19	and take
3	nearest value

n Distribution for

ts

2	3	4	5
20	34	22	8

Continuous Poisson D

Suppose X is an RV which accepts both integers & fractions then the corresponding PD is called Continuous PD.

There are two types

i, Normal Distribution

ii, Exponential DS

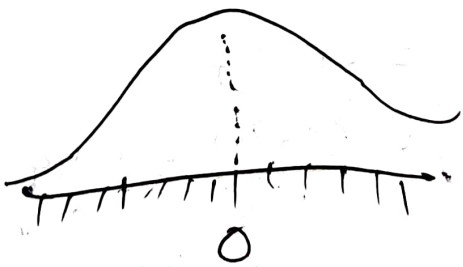
i Normal Distribution

Suppose X is a continuous RV with PF $\Rightarrow$ $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\left(\frac{x-\mu}{\sigma}\right)^2}$

then X follows Normal P

~~Bnorm~~ the graph

By using



Property

i, The curve is bell shaped

ii, Total area under normal curve is .

iii, A curve is symmetric about y axis

iv, Contains only 1 Mode

Exponential Distribution

Suppose X is a RV with DF

$f(x) = \lambda e^{-\lambda x}$ where λ is parameter

μ = mean, such that $\lambda = \frac{1}{\mu}$

Here $\sigma^2 = \frac{1}{\lambda^2} = \mu^2$

To find probability

$$P(X = x) = \int_a^b f(x)\,dx$$

$(a, b) \in [0, \infty]$

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2. The length of a telephone conversation

Poisson Distribution

Suppose X is a RV with P. mass function $P(X=R) = \frac{e^{-\lambda}\lambda^{r}}{r!}$ then

X follows Poisson Distribution

conditions for poisson distribution are

i, n is very large (or) experiment repeated many times

&

ii, For rarely happening events

iii, For low success rate

iv, If an event occurs infinitely in a specific time interval

In PDistribution $\mu = \sigma^2 = \lambda$